

Question Bank — DBMS & SQL

Module 3 companion | PYQ-style + Practice | exam-tagged, answers & explanations

■ How to use this bank

Two sections: PYQ-style (modelled on real IBPS/SSC/GATE/PSU/DSSSB questions, exam-tagged) and **Practice** (full syllabus coverage by sub-topic).

Every question has answer + explanation. **Tags:** IBPS SO IT • Coal India MT • DSSSB TGT • All 3 • GATE-level • SSC/other.

For IBPS: prioritise IBPS / All-3; GATE-tagged (relational algebra, normalization proofs, serializability) are Coal-India depth.

SECTION A — PYQ-style (modelled on actual exam questions)

Fundamentals & architecture

Q1

All 3 exams

A DBMS reduces which major problem of file-based systems?

- (a) Faster CPU
- (b) Data redundancy and inconsistency
- (c) Screen resolution
- (d) Cable length

Answer: (b)

A DBMS centralises data, cutting redundancy and the inconsistency it causes.

Q2

All 3 exams

SSC/other

The three-schema architecture consists of:

- (a) Input/Process/Output
- (b) External, Conceptual, Internal
- (c) Client/Server/DB
- (d) Primary/Secondary/Cache

Answer: (b)

ANSI-SPARC: External (views), Conceptual (logical), Internal (physical) levels.

Q3

All 3 exams

Physical data independence means changing the ____ without affecting the ____:

- (a) view; storage
- (b) storage; logical schema
- (c) logical; view
- (d) table; row

Answer: (b)

Physical independence = alter storage/internal schema without changing the conceptual/logical schema.

Q4

Coal India MT

GATE-level

Which language retrieves data without specifying how to get it (procedural vs non-procedural)?

- (a) Relational algebra (procedural)
- (b) Tuple relational calculus (non-procedural)
- (c) Assembly
- (d) C

Answer: (b)

Tuple relational calculus is non-procedural (declarative); relational algebra is procedural.

Keys

Q5

All 3 exams

A candidate key is best defined as:

- (a) Any set of attributes
- (b) A minimal super key
- (c) The first column
- (d) A foreign key

Answer: (b)

A candidate key is a minimal super key — no attribute can be removed while keeping uniqueness.

Q6

IBPS SO IT

All 3 exams

A primary key must be:

- (a) Nullable
- (b) Unique and NOT NULL
- (c) Composite always
- (d) Foreign

Answer: (b)

A primary key uniquely identifies a row and cannot be NULL.

Q7

All 3 exams

SSC/other

A foreign key enforces:

- (a) Domain constraint
- (b) Referential integrity
- (c) Entity integrity
- (d) Encryption

Answer: (b)

A foreign key references another table's primary key, enforcing referential integrity.

Q8

All 3 exams

The number of candidate keys a relation can have is:

- (a) Exactly one
- (b) At most one
- (c) One or more
- (d) Always two

Answer: (c)

A relation may have several candidate keys; one is chosen as primary, the rest are alternate keys.

Relational model & ER

Q9

All 3 exams

In relational terminology, a row is a:

- (a) Attribute
- (b) Tuple
- (c) Relation
- (d) Domain

Answer: (b)

A row = tuple; a column = attribute; a table = relation.

Q10

All 3 exams

SSC/other

The number of attributes in a relation is its:

- (a) Cardinality
- (b) Degree
- (c) Domain
- (d) Arity index

Answer: (b)

Degree = number of attributes (columns); cardinality = number of tuples (rows).

Q11

DSSSB TGT

All 3 exams

In an ER diagram, an entity is represented by a:

- (a) Ellipse
- (b) Rectangle
- (c) Diamond
- (d) Line

Answer: (b)

Entity = rectangle, attribute = ellipse, relationship = diamond.

Q12

All 3 exams

A relationship where one record relates to many is:

- (a) 1:1
- (b) 1:N
- (c) M:N
- (d) N:1 only

Answer: (b)

1:N (one-to-many) cardinality, e.g. one department has many employees.

Q13

DSSSB TGT

All 3 exams

A multivalued attribute in an ER diagram is shown by a:

- (a) Single ellipse
- (b) Double ellipse
- (c) Dashed ellipse
- (d) Rectangle

Answer: (b)

Double ellipse = multivalued (e.g. multiple phone numbers); dashed = derived.

Normalization

Q14

All 3 exams

First Normal Form (1NF) requires:

- (a) No partial dependency
- (b) Atomic values only
- (c) No transitive dependency
- (d) Super-key determinants

Answer: (b)

1NF = atomic (indivisible) values, no repeating groups/multivalued attributes.

Q15

IBPS SO IT

All 3 exams

Which normal form removes partial dependency?

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) BCNF

Answer: (b)

2NF removes partial dependency (non-key attribute depending on part of a composite key).

Q16

All 3 exams

3NF removes which type of dependency?

- (a) Partial
- (b) Transitive
- (c) Functional
- (d) Trivial

Answer: (b)

3NF removes transitive dependency (non-key attribute depending on another non-key attribute).

Q17

Coal India MT

GATE-level

BCNF requires that for every functional dependency $X \rightarrow Y$:

- (a) Y is a key
- (b) X is a super key
- (c) X is atomic
- (d) Y is prime

Answer: (b)

BCNF: the determinant X of every non-trivial FD must be a super key.

Q18

All 3 exams

SSC/other

Update, insertion and deletion anomalies are reduced by:

- (a) Indexing
- (b) Normalization
- (c) Encryption
- (d) Sharding

Answer: (b)

Normalization reorganises data to remove redundancy and the anomalies it causes.

Q19**GATE-level****Coal India MT**

The set of all attributes functionally determined by X is called:

- (a) X super key
- (b) Attribute closure X^+
- (c) Candidate set
- (d) Prime set

Answer: (b)

Attribute closure (X^+) is used to find candidate keys and test FDs. GATE-depth.

SQL

Q20**All 3 exams**

Which SQL category does CREATE TABLE belong to?

- (a) DML
- (b) DDL
- (c) DCL
- (d) TCL

Answer: (b)

CREATE (with ALTER, DROP, TRUNCATE) is DDL — Data Definition Language.

Q21**All 3 exams****SSC/other**

Which command permanently removes a table's structure and data?

- (a) DELETE
- (b) TRUNCATE
- (c) DROP
- (d) REMOVE

Answer: (c)

DROP removes the whole table (structure + data). DELETE removes rows; TRUNCATE removes all rows but keeps structure.

Q22**IBPS SO IT****All 3 exams**

Which clause filters groups after GROUP BY?

- (a) WHERE
- (b) HAVING
- (c) ORDER BY
- (d) FILTER

Answer: (b)

HAVING filters aggregated groups; WHERE filters rows before grouping.

Q23**All 3 exams**

Which JOIN returns only matching rows in both tables?

- (a) LEFT JOIN
- (b) RIGHT JOIN
- (c) INNER JOIN
- (d) FULL JOIN

Answer: (c)

INNER JOIN returns rows with matches in both tables.

Q24**DSSSB TGT****All 3 exams**

Which SQL function returns the number of rows?

- (a) SUM()
- (b) COUNT()
- (c) AVG()
- (d) LEN()

Answer: (b)

COUNT() returns the number of rows (or non-null values).

Q25**DSSSB TGT****All 3 exams**

The MySQL function UCASE()/UPPER() is used to:

- (a) Round a number
- (b) Convert text to uppercase
- (c) Get the date
- (d) Trim spaces

Answer: (b)

UCASE()/UPPER() converts a string to uppercase — a DSSSB DOE-listed function.

Q26**All 3 exams**

COMMIT and ROLLBACK belong to which SQL category?

- (a) DDL
- (b) DML
- (c) DCL
- (d) TCL

Answer: (d)

COMMIT/ROLLBACK/SAVEPOINT are TCL — Transaction Control Language.

Q27**GATE-level****Coal India MT**

In the logical query order, which executes first?

- (a) SELECT
- (b) FROM
- (c) WHERE
- (d) ORDER BY

Answer: (b)

Logical order: FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY.

Transactions & indexing

Q28**All 3 exams**

The 'A' in ACID stands for:

- (a) Availability
- (b) Atomicity
- (c) Access
- (d) Authentication

Answer: (b)

Atomicity — a transaction executes fully or not at all.

Q29

IBPS SO IT

All 3 exams

Which ACID property ensures committed data survives a crash?

- (a) Atomicity
- (b) Consistency
- (c) Isolation
- (d) Durability

Answer: (d)

Durability guarantees committed changes persist despite failures.

Q30

All 3 exams

SSC/other

Reading uncommitted data of another transaction is a:

- (a) Lost update
- (b) Dirty read
- (c) Phantom
- (d) Deadlock

Answer: (b)

A dirty read reads data written by an uncommitted (possibly-to-be-rolled-back) transaction.

Q31

Coal India MT

GATE-level

Two-Phase Locking (2PL) guarantees:

- (a) Deadlock freedom
- (b) Conflict serializability
- (c) No dirty reads always
- (d) Durability

Answer: (b)

2PL (growing then shrinking phase) guarantees conflict-serializable schedules (not deadlock freedom).

Q32

All 3 exams

An index primarily improves:

- (a) Insert speed
- (b) Read/search speed
- (c) Storage savings
- (d) Encryption

Answer: (b)

An index speeds up searches/reads (at the cost of extra storage and slower writes).

Q33

Coal India MT

GATE-level

Real databases typically implement indexes using:

- (a) Linked lists
- (b) B/B+ trees
- (c) Stacks
- (d) Hash of the whole table

Answer: (b)

B+ trees are standard for DB indexing — balanced, with linked leaves for range queries.

SECTION B — Practice (full syllabus coverage for revision)

B1. Fundamentals & keys

Q34

All 3 exams

A super key is:

- (a) Always minimal
- (b) Any attribute set that uniquely identifies a tuple
- (c) A foreign key
- (d) Always one column

Answer: (b)

A super key uniquely identifies a tuple but may contain extra attributes; a candidate key is a minimal super key.

Q35

All 3 exams

Entity integrity means:

- (a) Foreign keys match
- (b) Primary key is not NULL
- (c) Data is encrypted
- (d) Rows are sorted

Answer: (b)

Entity integrity: no part of a primary key may be NULL.

Q36

All 3 exams

Which level of the 3-schema architecture do end users see?

- (a) Internal
- (b) Conceptual
- (c) External (view)
- (d) Physical

Answer: (c)

Users interact with the external/view level, customised per user.

Q37

SSC/other

All 3 exams

The DBMS component that ensures data survives failures is:

- (a) Query optimizer
- (b) Recovery manager
- (c) Parser
- (d) Scheduler

Answer: (b)

The recovery manager restores a consistent state after crashes (logs, checkpoints).

B2. Relational model & ER

Q38

All 3 exams

A domain is:

- (a) A row
- (b) The set of allowed values for an attribute
- (c) A table
- (d) A key

Answer: (b)

A domain defines the permissible values for an attribute.

Q39

All 3 exams

A weak entity:

- (a) Has its own key
- (b) Depends on an owner entity for identification
- (c) Has no attributes
- (d) Is a relationship

Answer: (b)

A weak entity cannot be uniquely identified by its own attributes; it needs an owner (identifying) entity.

Q40

DSSSB TGT

All 3 exams

A derived attribute (e.g. age from DOB) is drawn as a:

- (a) Double ellipse
- (b) Dashed ellipse
- (c) Rectangle
- (d) Diamond

Answer: (b)

Derived attributes use a dashed ellipse.

Q41

GATE-level

Coal India MT

Converting an M:N relationship to tables requires:

- (a) No extra table
- (b) A separate junction/bridge table
- (c) Merging both entities
- (d) A view

Answer: (b)

An M:N relationship needs a separate relationship (junction) table with both foreign keys.

B3. Normalization

Q42

All 3 exams

A table is in 2NF if it is in 1NF and:

- (a) Has no transitive dependency
- (b) Has no partial dependency
- (c) Determinants are super keys
- (d) Has an index

Answer: (b)

2NF = 1NF + no partial dependency on a composite key.

Q43

All 3 exams

Which anomaly means deleting one fact loses another?

- (a) Insertion
- (b) Update
- (c) Deletion
- (d) Read

Answer: (c)

A deletion anomaly loses unrelated data when a row is removed.

Q44**GATE-level****Coal India MT**

A decomposition is lossless if the natural join of the parts yields:

- (a) More rows
- (b) Exactly the original relation
- (c) Fewer columns
- (d) A view

Answer: (b)

Lossless-join decomposition reconstructs exactly the original relation with no spurious tuples.

Q45**GATE-level****Coal India MT**

4NF deals with:

- (a) Partial dependency
- (b) Transitive dependency
- (c) Multivalued dependency
- (d) Foreign keys

Answer: (c)

4NF removes non-trivial multivalued dependencies.

B4. SQL

Q46**All 3 exams**

Which keyword removes duplicate rows from a result?

- (a) UNIQUE
- (b) DISTINCT
- (c) ONLY
- (d) REMOVE

Answer: (b)

SELECT DISTINCT removes duplicate rows from the output.

Q47**All 3 exams**

A LEFT JOIN returns:

- (a) Only matches
- (b) All left rows + matching right (NULLs if none)
- (c) All right rows
- (d) Cartesian product

Answer: (b)

LEFT OUTER JOIN keeps all left-table rows, filling NULLs where the right has no match.

Q48**All 3 exams**

Which aggregate ignores NULLs when averaging?

- (a) SUM
- (b) AVG
- (c) COUNT(*)
- (d) MAX

Answer: (b)

AVG (and most aggregates) ignore NULLs; COUNT(*) counts all rows including NULLs.

Q49**DSSSB TGT****All 3 exams****MySQL MOD(10,3) returns:**

- (a) 3
- (b) 1
- (c) 0
- (d) 10

Answer: (b)MOD returns the remainder: $10 \bmod 3 = 1$.**Q50****All 3 exams****The wildcard for 'any sequence of characters' in SQL LIKE is:**

- (a) _
- (b) %
- (c) *
- (d) ?

Answer: (b)

In LIKE, % matches any sequence; _ matches a single character.

Q51**All 3 exams****SSC/other****TRUNCATE differs from DELETE in that TRUNCATE:**

- (a) Can use WHERE
- (b) Removes all rows and is faster, hard to roll back
- (c) Removes structure
- (d) Is DML

Answer: (b)

TRUNCATE (DDL) quickly removes all rows, cannot use WHERE, and is not easily rolled back.

B5. Transactions & indexing

Q52**All 3 exams****The 'C' in ACID ensures a transaction moves the DB between:**

- (a) Two servers
- (b) Valid (consistent) states
- (c) Two users
- (d) Cache and disk

Answer: (b)

Consistency: a transaction takes the database from one valid state to another.

Q53**All 3 exams****Isolation ensures concurrent transactions:**

- (a) Run faster
- (b) Do not interfere; result equals some serial order
- (c) Share memory
- (d) Skip logging

Answer: (b)

Isolation makes concurrent execution equivalent to some serial schedule.

Q54

All 3 exams

A dense index has:

- (a) One entry per block
- (b) One entry per record
- (c) No entries
- (d) Entries only for keys

Answer: (b)

A dense index has an entry for every record; a sparse index has one per block.

Q55

GATE-level

Coal India MT

A schedule is conflict-serializable if its precedence graph:

- (a) Has a cycle
- (b) Is acyclic
- (c) Is empty
- (d) Has one node

Answer: (b)

Conflict-serializable $\Leftrightarrow$ the precedence (serialization) graph is acyclic.

Q56

All 3 exams

Compared to a B-tree, a B+ tree stores data pointers:

- (a) In all nodes
- (b) Only in leaf nodes (linked)
- (c) Only in the root
- (d) Nowhere

Answer: (b)

B+ trees keep all data in linked leaves, making range scans efficient.

Coverage & next

✓ What this bank covered

PYQ + Practice across DBMS sub-topics: fundamentals/architecture, keys, relational model & ER, normalization, SQL (DDL/DML/joins/functions), transactions/ACID, indexing.

GATE-tagged items (attribute closure, lossless join, 4NF, conflict-serializability, 2PL) are Coal-India depth.

Note: original questions modelled on real exam patterns, not verbatim reproductions.